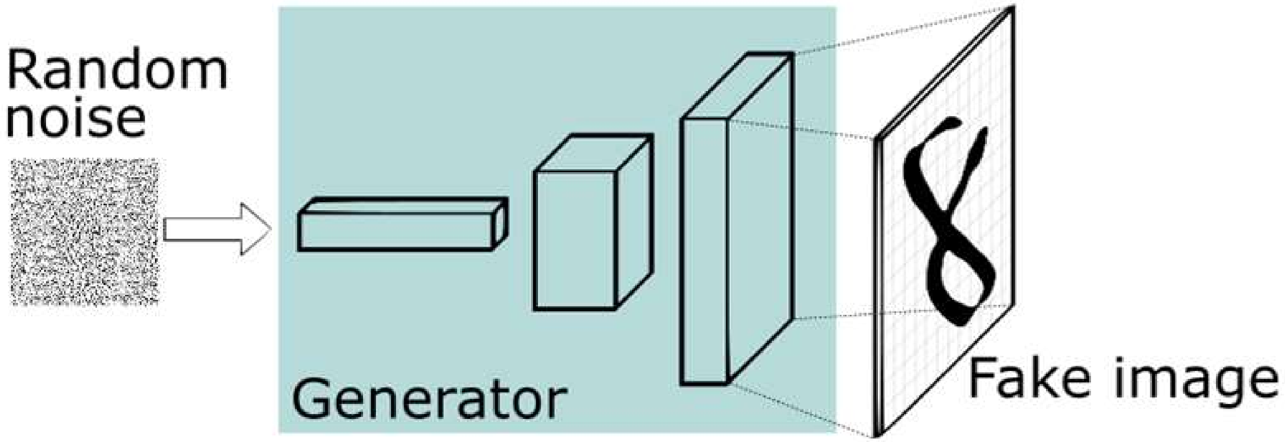
GENERATIVE MODELS

COMP 4107

WHY GENERATIVE MODELS?

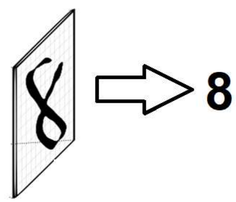
- Synthetic data



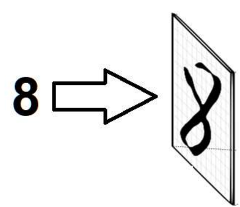
Source: https://pathmind.com/wiki/generative-adversarial-network-gan

WHAT DOES "GENERATIVE" MEAN?

- Discriminative
  - Identify labels from features



- Generative
  - Identify feature distribution from labels



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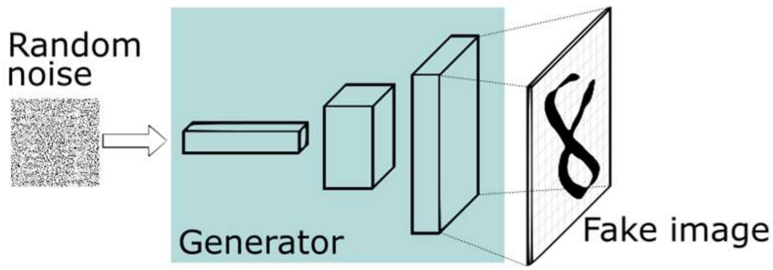
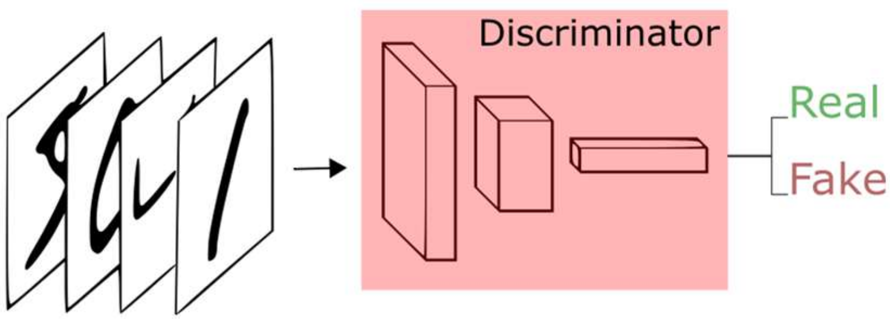
GENERATIVE MODELS SO FAR?

GAN ARCHITECTURE

- Generator



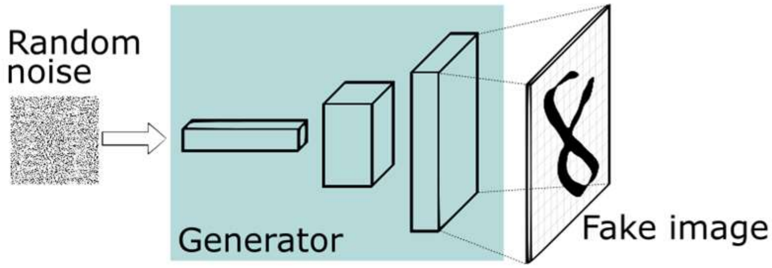
- Discriminator



Source: https://pathmind.com/wiki/generative-adversarial-network-gan

GAN GENERATOR

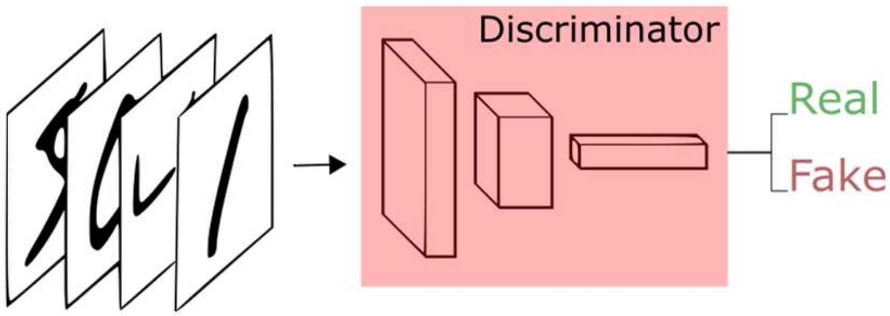
- Input: Random noise vector
- Output: Fake image



Source: https://pathmind.com/wiki/generative-adversarial-network-gan

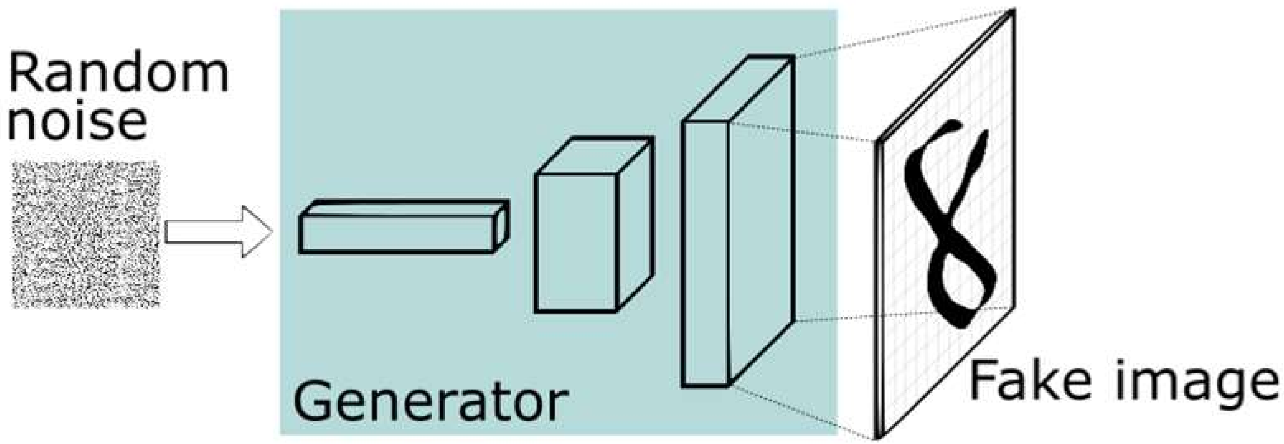
GAN DISCRIMINATOR

- Input: Images
- Output: Real or Fake



Source: https://pathmind.com/wiki/generative-adversarial-network-gan

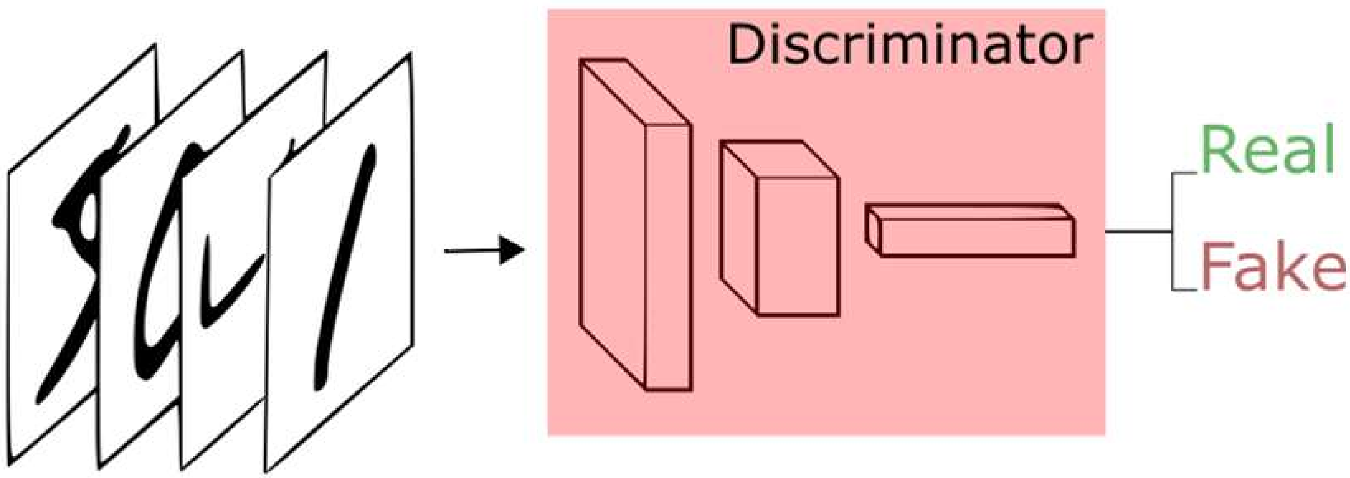
GENERATOR TRAINING



$$L(x_i) = -\log\left(D\left(G(x_i)\right)\right)$$

Source: https://pathmind.com/wiki/generative-adversarial-network-gan

DISCRIMINATOR TRAINING

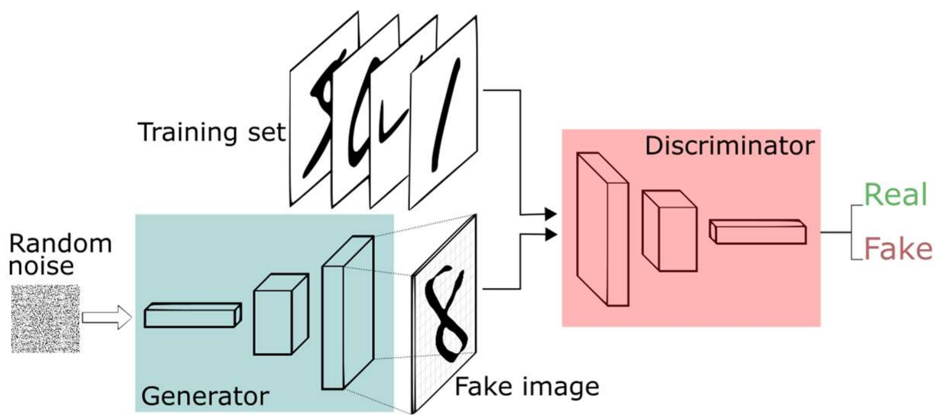


$$L\big((x_i, y_i)\big) = -y_i \log\big(D(x_i)\big) - (1 - y_i)\log\big(1 - D(x_i)\big)$$

Source: https://pathmind.com/wiki/generative-adversarial-network-gan

GAN TRAINING

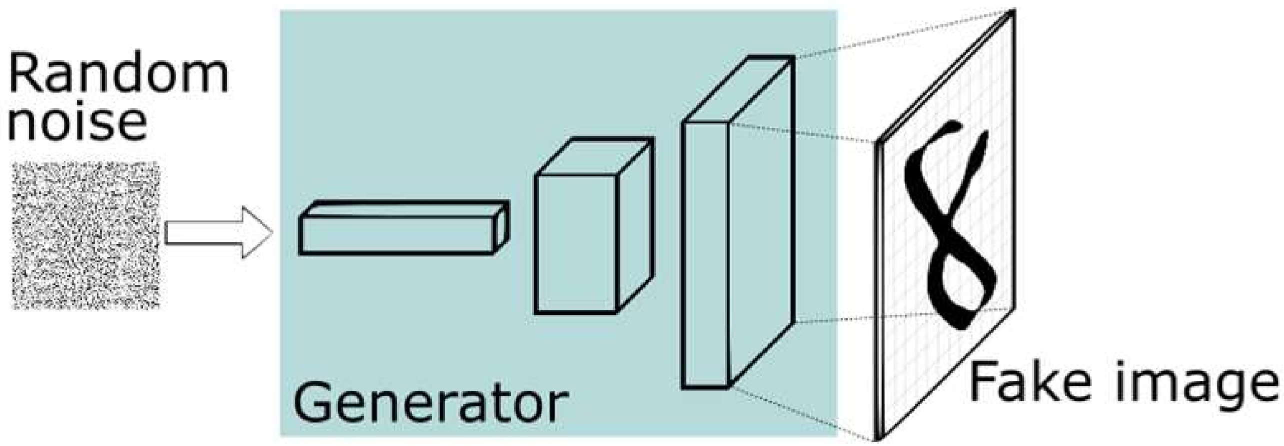
1. Use generator to generate a set of fake images.
2. Classify images as real or fake.
3. Update discriminator with back-propagation.
4. Use generator to generate a set of fake images.
5. Update generator with back-propagation.
6. Repeat.



Source: https://pathmind.com/wiki/generative-adversarial-network-gan

WHAT DO WE GET?

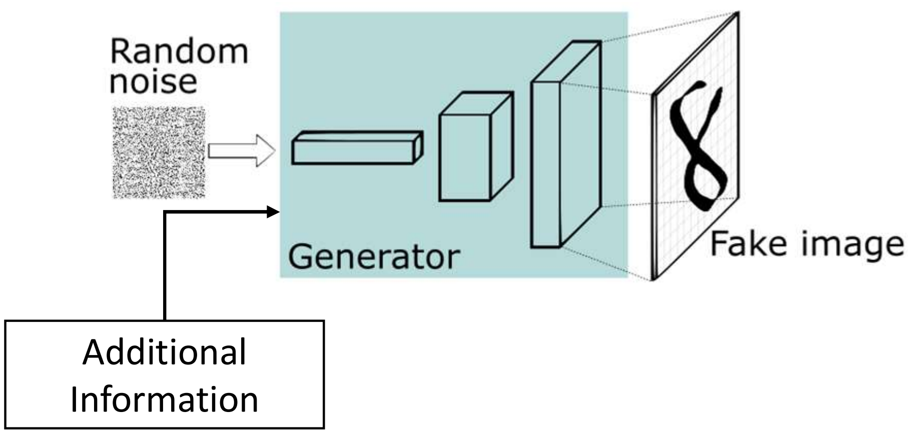
- Generator generates realistic fake images

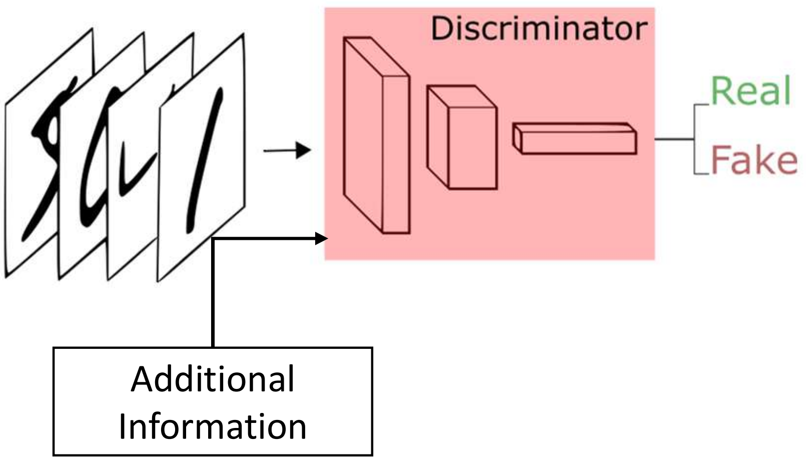


Source: https://pathmind.com/wiki/generative-adversarial-network-gan

CONDITIONAL GANS

- Guides the generator
  - Generate specific types of images
  - More stable training





Source: https://pathmind.com/wiki/generative-adversarial-network-gan

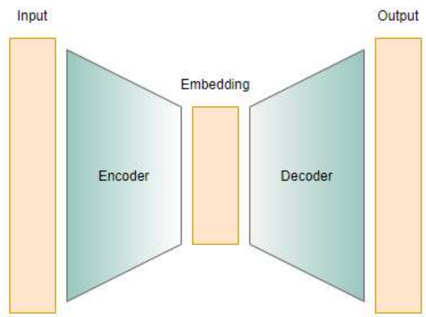
HOW DO WE USE THIS?

Task: Create an image classifier

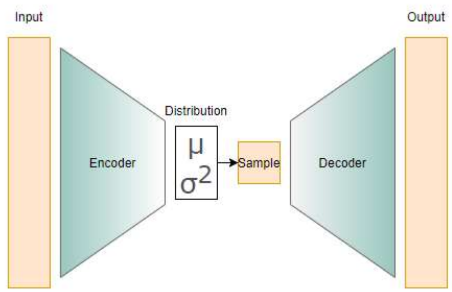
1. Train GAN using original training set.
2. Use generator to create realistic fake images.
3. Train classifier on augmentation dataset.

VARIATIONAL AUTOENCODERS

- Autoencoder

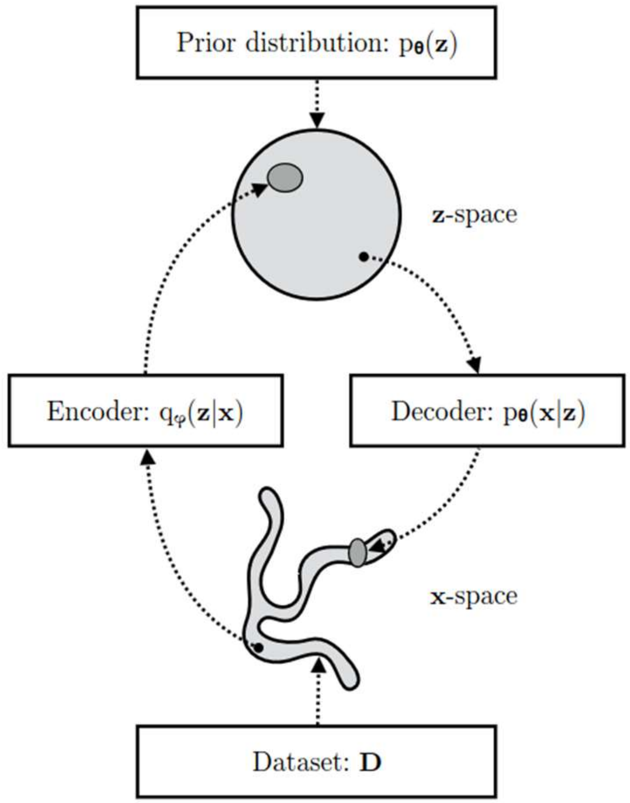


- Variational Autoencoder



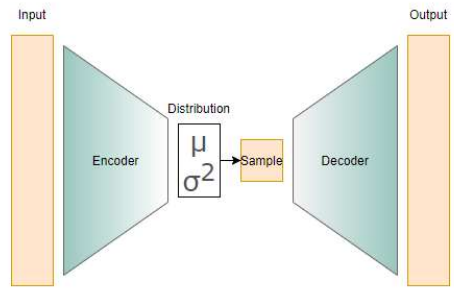
Source: https://www.andrew.cmu.edu/user/xihuang/blog/ae_vae/ae_vae.html

VARIATIONAL AUTOENCODERS



Source: https://arxiv.org/pdf/1906.02691.pdf

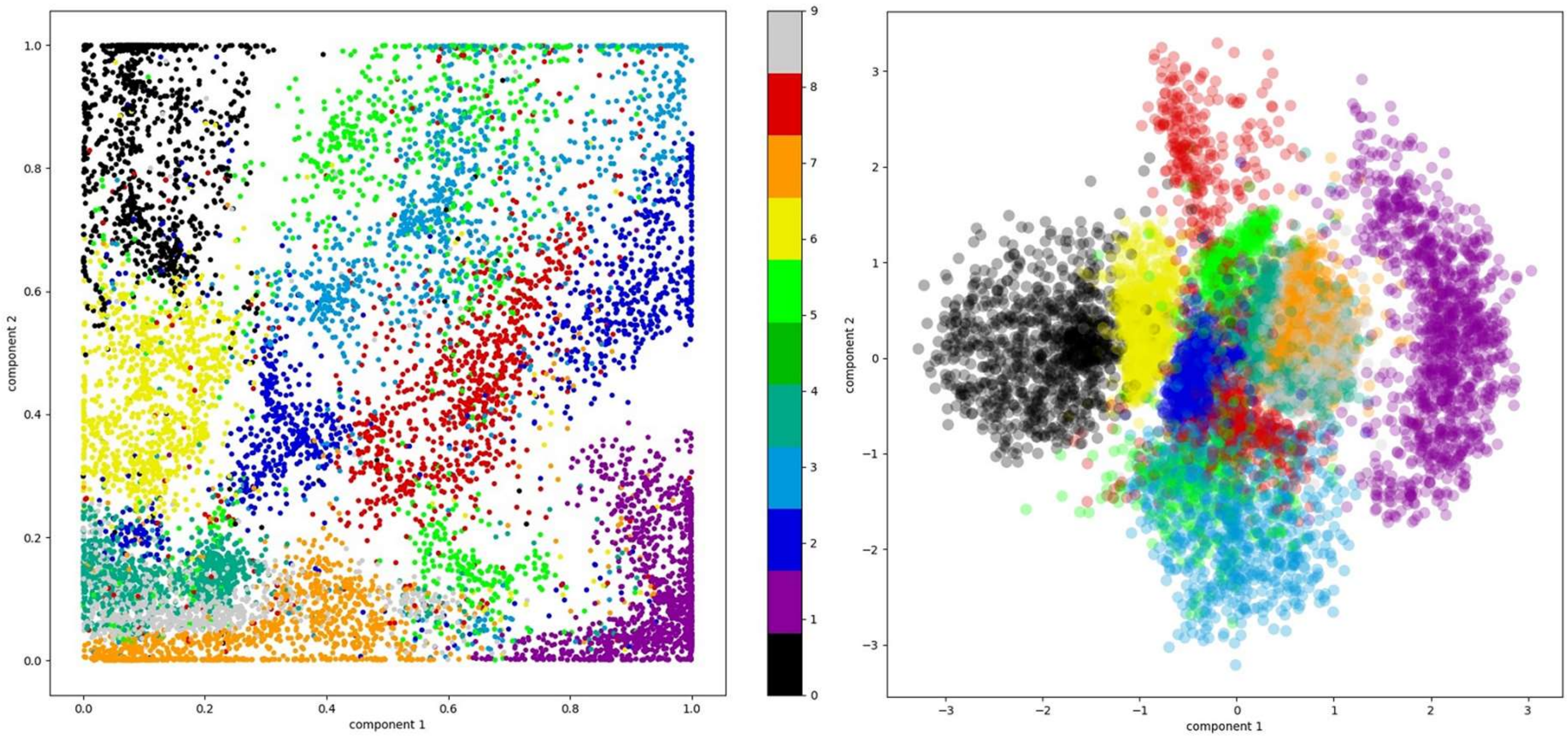
TRAINING VAES



$$ReconstructionLoss = |x - \hat{x}|$$
$$SimilarityLoss = D_{KL}(N(\mu, \sigma), N(0, 1))$$
$$Loss = ReconstructionLoss + SimilarityLoss$$

Source: https://www.andrew.cmu.edu/user/xihuang/blog/ae_vae/ae_vae.html

LATENT DISTRIBUTION



Source: https://pureai.com/articles/2020/05/07/variational-autoencoders.aspx

REFERENCES

- Mirz & Osindero, Conditional Generative Adversarial Nets (https://arxiv.org/abs/1411.1784)
- Goodfellow et al., Generative Adversarial Networks (https://arxiv.org/abs/1406.2661)
- Goodfellow, Bengio & Courville, Deep Learning Book Chapter 20 (https://www.deeplearningbook.org/contents/generative_models.html)
- Kingma & Welling, An Introduction to Variational Autoencoders (https://arxiv.org/pdf/1906.02691.pdf)